Week 3, Lecture 2
Squeezed states of light

Lecturer: Josephine
Email: j.dias@uq.edu.au
Office: Parnell Room 320
Office hours: Thursdays 3-4pm

Notes 5
Bachor & Ralph, 9.1, 9.2
Walls & Milburn, 1.4, 1.5

Review

What we covered last lecture…

We covered three different quasi-probability distributions used in quantum optics:

Husimi Q function

Q function can be thought of as a probability density
(P and W functions cannot)

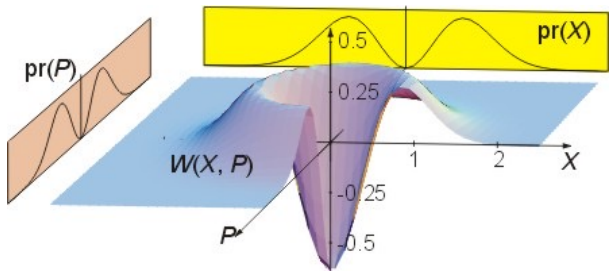
Glauber Sudarshan P function

P function can be negative and singular

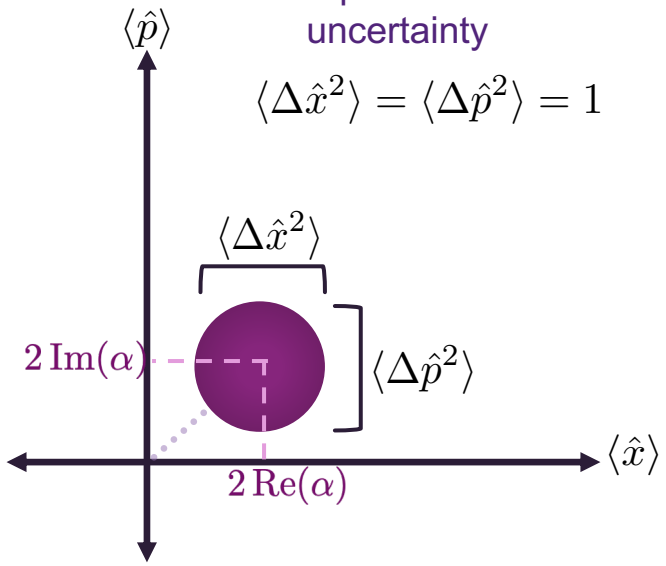
Wigner function

W function can be negative

Wigner function negativity is a sign of non-classicality e.g. $|1\rangle$



Coherent states - review



Coherent state

Uncertainty principle:

$$\langle\Delta\hat{x}^2\rangle\,\langle\Delta\hat{p}^2\rangle \geq \frac{\hbar^2}{4}$$

Generalised quadrature operators

$\hbar = 2$ ➡

$$\hat{x}_\theta = \hat{a}e^{-i\theta} + \hat{a}^\dagger e^{i\theta}$$
$$\hat{x}_{\theta+\pi/2} = i(\hat{a}^\dagger e^{i\theta} - \hat{a}e^{-i\theta})$$

$\theta = 0$ ➡

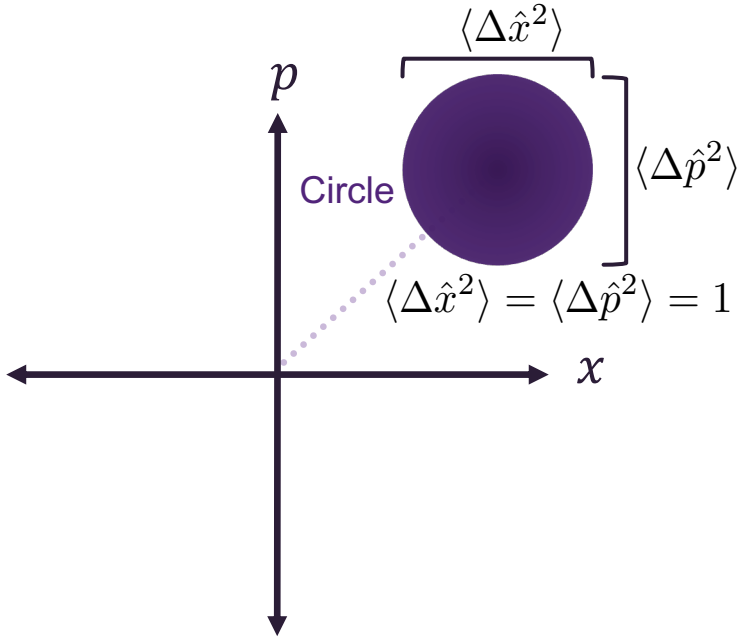
$$\hat{x} = \hat{a} + \hat{a}^\dagger$$
$$\hat{p} = i(\hat{a}^\dagger - \hat{a})$$

$$\langle\Delta\hat{x}^2\rangle\,\langle\Delta\hat{p}^2\rangle \geq 1$$

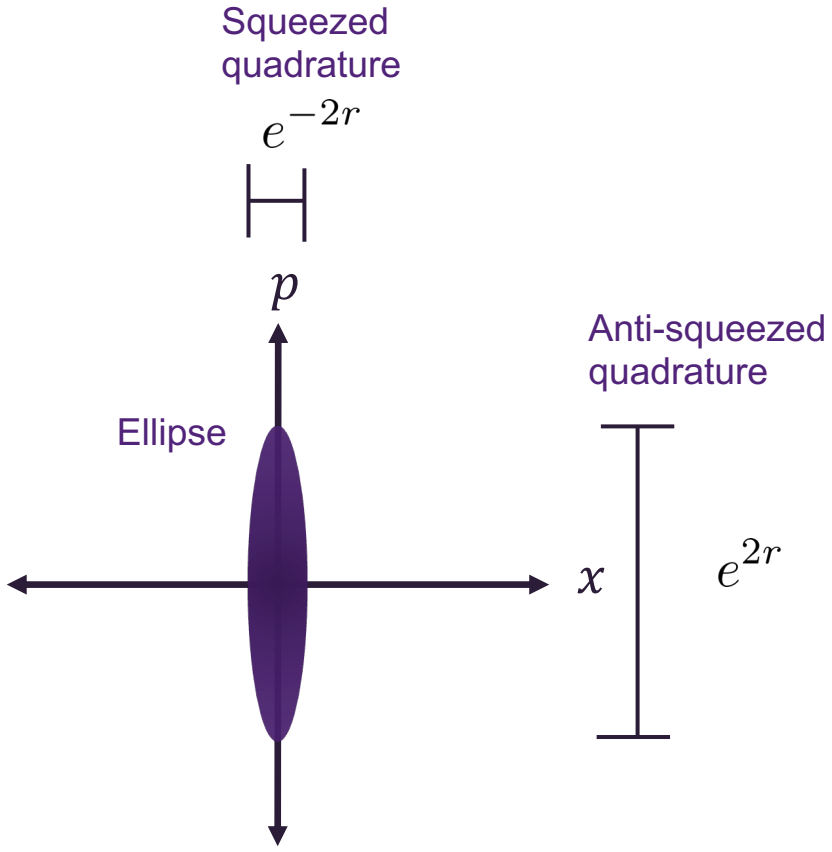
Coherent states are minimum uncertainty states

$$\langle\Delta\hat{x}^2\rangle\,\langle\Delta\hat{p}^2\rangle = 1$$

Squeezed states



Coherent state



Squeezed state

Coherent states and squeezed states are minimum uncertainty states

$$\langle \Delta\hat{x}^2 \rangle \langle \Delta\hat{p}^2 \rangle = 1$$

Asymmetry in noise distribution: can have either $\langle \Delta\hat{x}^2 \rangle < 1$ or $\langle \Delta\hat{p}^2 \rangle < 1$ but not both

See chapter 5 of the lecture notes for squeezing

Also chapter 9 of Bachor, Hans-A. / Ralph, Timothy C.
A guide to experiments in quantum optics (2019)

Squeezing operator

The squeezing operator is defined as:

$$S(\zeta) = \exp\left(\left(\zeta^* \hat{a}^2 - \zeta \hat{a}^{\dagger 2}\right)/2\right)$$

Complex squeezing parameter

$$\zeta = re^{2i\phi}$$

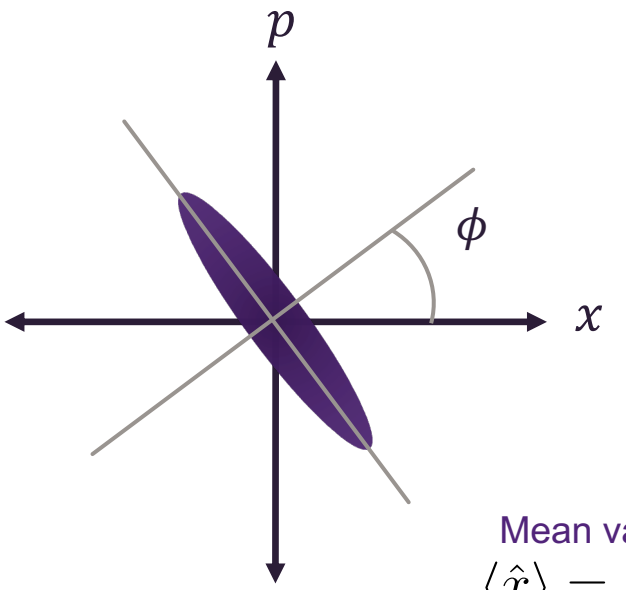
r controls the strength of squeezing
ϕ controls the angle of squeezing

$\phi = 0$ corresponds to x squeezing

Apply the squeezing operator to the vacuum state:

$$|\zeta\rangle = \hat{S}(\zeta)\,|0\rangle$$

Squeezed vacuum state



Mean value

$$\langle \hat{x} \rangle = \langle \hat{p} \rangle = 0$$

Ellipse lies at origin

Squeezing operator properties

Useful identities

$$S(\zeta) = \exp\left(\left(\zeta^* \hat{a}^2 - \zeta \hat{a}^{\dagger 2}\right)/2\right)$$

Squeezing operator is unitary

$$\begin{aligned}
\hat{S}^\dagger(\zeta) &= \hat{S}^{-1}(\zeta) = \hat{S}(-\zeta) \\
\hat{S}(\zeta) a \hat{S}^\dagger(\zeta) &= \mu a + \nu a^\dagger \\
\hat{S}(\zeta) a^\dagger \hat{S}^\dagger(\zeta) &= \mu a^\dagger + \nu^* a \\
\hat{S}^\dagger(\zeta) a \hat{S}(\zeta) &= \mu a - \nu a^\dagger \\
\hat{S}^\dagger(\zeta) a^\dagger \hat{S}(\zeta) &= \mu a^\dagger - \nu^* a
\end{aligned}$$

$$\mu = \cosh r \qquad \nu = e^{2i\phi} \sinh r \qquad \mu^2 - |\nu|^2 = 1$$

Squeezed vacuum state

Variance

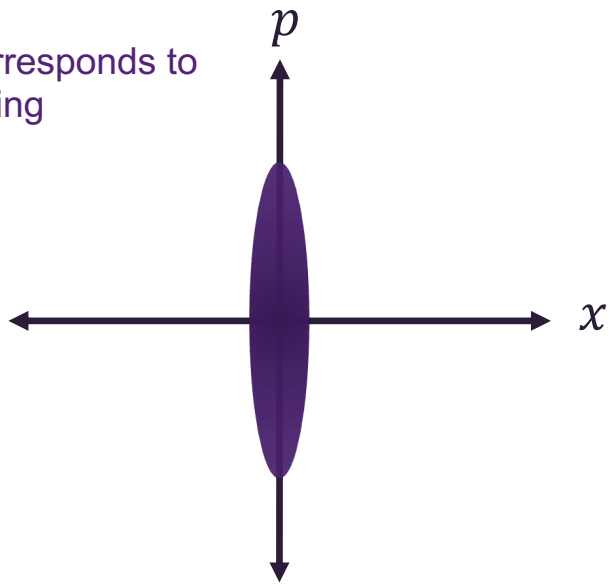
$$S(\zeta) = \exp\left(\left(\zeta^* \hat{a}^2 - \zeta \hat{a}^{\dagger 2}\right)/2\right)$$

$$\zeta = re^{2i\phi} \qquad \phi = 0 \Longrightarrow \qquad \zeta = r$$

$$S(r) = \exp\left(\left(r\hat{a}^2 - r\hat{a}^{\dagger 2}\right)/2\right)$$

$$|r\rangle = \hat{S}(r)\,|0\rangle$$

$\phi = 0$ corresponds to x squeezing



Example: calculate the variance of x for the squeezed vacuum state where $\phi = 0$

$$\mathrm{var}(x) = \langle \Delta \hat{x}^2 \rangle = \langle \hat{x}^2 \rangle - \langle \hat{x} \rangle^2 \qquad \hat{x} = \hat{a} + \hat{a}^\dagger$$

Squeezed vacuum state

Variance

$$\langle \hat{x} \rangle = \langle 0 | \, \hat{S}^\dagger(r) \left(\hat{a} + \hat{a}^\dagger \right) \hat{S}(r) \, | 0 \rangle$$

$$\langle \hat{x} \rangle = \langle 0 | \left[\hat{S}^\dagger(r) \hat{a} \hat{S}(r) + \hat{S}^\dagger(r) \hat{a}^\dagger \hat{S}(r) \right] | 0 \rangle$$

$$\underbrace{\hat{S}^\dagger(r) \hat{a} \hat{S}(r)}_{\mu \hat{a} - \nu \hat{a}^\dagger} \quad \underbrace{\hat{S}^\dagger(r) \hat{a}^\dagger \hat{S}(r)}_{\mu \hat{a}^\dagger - \nu \hat{a}}$$

$$\langle \hat{x} \rangle = \langle 0 | \left[\mu \hat{a} - \nu \hat{a}^\dagger + \mu \hat{a}^\dagger - \nu \hat{a} \right] | 0 \rangle$$

$$\langle \hat{x} \rangle = 0$$

For the vacuum state, $\langle \hat{x} \rangle = 0$

Squeezing doesn't change the mean value of x

$$\hat{S}^\dagger(\zeta) a \hat{S}(\zeta) = \mu a - \nu a^\dagger$$

$$\hat{S}^\dagger(\zeta) a^\dagger \hat{S}(\zeta) = \mu a^\dagger - \nu^* a$$

$$\mu = \cosh r$$

$$\nu = \sinh r$$

Squeezed vacuum state

Variance

$$\langle \hat{x}^2 \rangle = \langle 0| \hat{S}^\dagger(r) \left(\hat{a} + \hat{a}^\dagger\right)^2 \hat{S}(r) |0\rangle$$

$$\langle \hat{x}^2 \rangle = \langle 0| \hat{S}^\dagger(r) \left[\hat{a}\hat{a} + \hat{a}^\dagger\hat{a}^\dagger + \hat{a}^\dagger\hat{a} + \hat{a}\hat{a}^\dagger\right] \hat{S}(r) |0\rangle$$

$$[\hat{a}, \hat{a}^\dagger] = 1 \qquad \hat{a}\hat{a}^\dagger = 1 + \hat{a}^\dagger\hat{a}$$

$$\langle \hat{x}^2 \rangle = \langle 0| \hat{S}^\dagger(r) \left[\hat{a}\hat{a} + \hat{a}^\dagger\hat{a}^\dagger + 2\hat{a}^\dagger\hat{a} + 1\right] \hat{S}(r) |0\rangle$$

$$\langle \hat{x}^2 \rangle = \langle 0| \left[\hat{S}^\dagger(r)\hat{a}\hat{a}\hat{S}(r) + \hat{S}^\dagger(r)\hat{a}^\dagger\hat{a}^\dagger\hat{S}(r) + 2\hat{S}^\dagger(r)\hat{a}^\dagger\hat{a}\hat{S}(r) + \hat{S}^\dagger(r)\hat{S}(r)\right] |0\rangle$$

insert identity $\hat{S}\hat{S}^\dagger$

squeezing operator is unitary $\hat{S}(r)^\dagger\hat{S}(r) = \mathbb{1}$

$$\langle \hat{x}^2 \rangle = \langle 0| \left[\hat{S}^\dagger(r)\hat{a}\hat{S}(r)\hat{S}^\dagger(r)\hat{a}\hat{S}(r) + \hat{S}^\dagger(r)\hat{a}^\dagger\hat{S}(r)\hat{S}^\dagger(r)\hat{a}^\dagger\hat{S}(r) + 2\hat{S}^\dagger(r)\hat{a}^\dagger\hat{S}(r)\hat{S}^\dagger(r)\hat{a}\hat{S}(r) + 1\right] |0\rangle$$

$$\hat{S}^\dagger(\zeta) a \hat{S}(\zeta) = \mu a - \nu a^\dagger$$

$$\hat{S}^\dagger(\zeta) a^\dagger \hat{S}(\zeta) = \mu a^\dagger - \nu^* a$$

$$\mu = \cosh r$$

$$\nu = \sinh r$$

Squeezed vacuum state

Variance

$$\langle \hat{x}^2 \rangle = \langle 0| \Big[\underbrace{\hat{S}^\dagger(r)\hat{a}\hat{S}(r)}_{\mu\hat{a}-\nu\hat{a}^\dagger} \hat{S}^\dagger(r)\hat{a}\hat{S}(r) + \underbrace{\hat{S}^\dagger(r)\hat{a}^\dagger\hat{S}(r)}_{\mu\hat{a}^\dagger-\nu\hat{a}} \hat{S}^\dagger(r)\hat{a}^\dagger\hat{S}(r) + 2\hat{S}^\dagger(r)\hat{a}^\dagger\hat{S}(r)\hat{S}^\dagger(r)\hat{a}\hat{S}(r) + 1 \Big] |0\rangle$$

$$\langle \hat{x}^2 \rangle = \langle 0| \left[(\mu\hat{a} - \nu\hat{a}^\dagger)(\mu\hat{a} - \nu\hat{a}^\dagger) + (\mu\hat{a}^\dagger - \nu\hat{a})(\mu\hat{a}^\dagger - \nu\hat{a}) + 2(\mu\hat{a}^\dagger - \nu\hat{a})(\mu\hat{a} - \nu\hat{a}^\dagger) + 1 \right] |0\rangle$$

$$\langle 0|\hat{a}\hat{a}|0\rangle = 0 \qquad \langle 0|\hat{a}^\dagger\hat{a}^\dagger|0\rangle = 0 \qquad \hat{a}\,|0\rangle = 0$$

only keep $\hat{a}\hat{a}^\dagger$ (antinormally ordered) terms

$$\langle \hat{x}^2 \rangle = \langle 0| \left[-\mu\nu\hat{a}\hat{a}^\dagger - \nu\mu\hat{a}\hat{a}^\dagger + 2\nu^2\hat{a}\hat{a}^\dagger + 1 \right] |0\rangle$$

$$\mu = \cosh r \qquad \nu = \sinh r$$

hyperbolic trig identities

$$\cosh^2 r - \sinh^2 r = 1$$

$$\langle \hat{x}^2 \rangle = -\mu\nu - \nu\mu + 2\nu^2 + \mu^2 - \nu^2$$

Squeezed vacuum state

Variance

$$\langle \hat{x}^2 \rangle = -\mu\nu - \nu\mu + 2\nu^2 + \mu^2 - \nu^2$$

$$\langle \hat{x}^2 \rangle = -\mu\nu - \nu\mu + \mu^2 + \nu^2$$

$$\langle \hat{x}^2 \rangle = (\mu - \nu)^2$$

$$\mu = \cosh r \qquad \nu = \sinh r$$

$$\langle \hat{x}^2 \rangle = \underbrace{(\cosh r - \sinh r)^2}_{e^{-2r}}$$

$$\langle \hat{x}^2 \rangle = e^{-2r}$$

$$\mathrm{var}\,(x) = \langle \Delta \hat{x}^2 \rangle = \underbrace{\langle \hat{x}^2 \rangle}_{e^{-2r}} - \underbrace{\langle \hat{x} \rangle^2}_{0} = e^{-2r}$$

For the vacuum state: $\mathrm{var}\,(x) = 1$

After applying squeezing operator to vacuum: $\mathrm{var}\,(x) = e^{-2r} < 1$

Squeezed vacuum state

Variance

Next: calculate the variance of p for the squeezed vacuum state

$$\mathrm{var}\,(p) = \langle \Delta\hat{p}^2 \rangle = \langle \hat{p}^2 \rangle - \langle \hat{p} \rangle^2 \qquad\qquad \hat{p} = i(\hat{a}^\dagger - \hat{a})$$

$$\langle \hat{p} \rangle = \langle 0 | \hat{S}^\dagger(r) i \left(\hat{a}^\dagger - \hat{a} \right) \hat{S}(r) | 0 \rangle$$

$$\langle \hat{p} \rangle = i \langle 0 | \Big(\underbrace{\hat{S}^\dagger(r) \hat{a}^\dagger \hat{S}(r)}_{\mu\hat{a}^\dagger - \nu\hat{a}} - \underbrace{\hat{S}^\dagger(r) \hat{a} \hat{S}(r)}_{\mu\hat{a} - \nu\hat{a}^\dagger} \Big) | 0 \rangle$$

$$\langle \hat{p} \rangle = i \langle 0 | \left(\mu\hat{a}^\dagger - \nu\hat{a} + \mu\hat{a} - \nu\hat{a}^\dagger \right) | 0 \rangle$$

$$\langle \hat{p} \rangle = 0$$

For the vacuum state, $\langle \hat{p} \rangle = 0$

Squeezing doesn't change the mean value of p

Squeezed vacuum state

Variance

$$\hat{p} = i(\hat{a}^\dagger - \hat{a})$$

$$\langle \hat{p}^2 \rangle = -\langle 0| \hat{S}^\dagger(r) \left(\hat{a}^\dagger - \hat{a}\right)^2 \hat{S}(r) |0\rangle$$

$$\langle \hat{p}^2 \rangle = -\langle 0| \hat{S}^\dagger(r) \left[\hat{a}^\dagger\hat{a}^\dagger + \hat{a}\hat{a} - \hat{a}\hat{a}^\dagger - \hat{a}^\dagger\hat{a}\right] \hat{S}(r) |0\rangle$$

$$[\hat{a}, \hat{a}^\dagger] = 1 \qquad \hat{a}\hat{a}^\dagger = 1 + \hat{a}^\dagger\hat{a}$$

$$\langle \hat{p}^2 \rangle = -\langle 0| \hat{S}^\dagger(r) \left[\hat{a}^\dagger\hat{a}^\dagger + \hat{a}\hat{a} - 1 - 2\hat{a}^\dagger\hat{a}\right] \hat{S}(r) |0\rangle$$

$$\langle \hat{p}^2 \rangle = -\langle 0| \left[\hat{S}^\dagger(r)\hat{a}^\dagger\hat{a}^\dagger\hat{S}(r) + \hat{S}^\dagger(r)\hat{a}\hat{a}\hat{S}(r) - 2\hat{S}^\dagger(r)\hat{a}^\dagger\hat{a}\hat{S}(r) - \hat{S}^\dagger(r)\hat{S}(r)\right] |0\rangle$$

$$\langle \hat{p}^2 \rangle = -\langle 0| \Big[\underbrace{\hat{S}^\dagger(r)\hat{a}^\dagger\hat{S}(r)}_{\mu\hat{a}^\dagger - \nu\hat{a}}\hat{S}^\dagger(r)\hat{a}^\dagger\hat{S}(r) + \underbrace{\hat{S}^\dagger(r)\hat{a}\hat{S}(r)}_{\mu\hat{a} - \nu\hat{a}^\dagger}\hat{S}^\dagger(r)\hat{a}\hat{S}(r) - 2\hat{S}^\dagger(r)\hat{a}^\dagger\hat{S}(r)\hat{S}^\dagger(r)\hat{a}\hat{S}(r) - 1\Big] |0\rangle$$

Squeezed vacuum state

Variance

$$\langle \hat{p}^2 \rangle = -\langle 0| \Big[\underbrace{\hat{S}^\dagger(r)\hat{a}^\dagger \hat{S}(r)}_{\mu\hat{a}^\dagger - \nu\hat{a}} \hat{S}^\dagger(r)\hat{a}^\dagger \hat{S}(r) + \underbrace{\hat{S}^\dagger(r)\hat{a}\hat{S}(r)}_{\mu\hat{a} - \nu\hat{a}^\dagger} \hat{S}^\dagger(r)\hat{a}\hat{S}(r) - 2\hat{S}^\dagger(r)\hat{a}^\dagger \hat{S}(r)\hat{S}^\dagger(r)\hat{a}\hat{S}(r) - 1 \Big] |0\rangle$$

$$\langle \hat{p}^2 \rangle = -\langle 0| \left[(\mu\hat{a}^\dagger - \nu\hat{a})(\mu\hat{a}^\dagger - \nu\hat{a}) + (\mu\hat{a} - \nu\hat{a}^\dagger)(\mu\hat{a} - \nu\hat{a}^\dagger) - 2(\mu\hat{a}^\dagger - \nu\hat{a})(\mu\hat{a} - \nu\hat{a}^\dagger) - 1 \right] |0\rangle$$

only keep $\hat{a}\hat{a}^\dagger$ (antinormally ordered) terms $\quad \langle 0|\hat{a}\hat{a}|0\rangle = 0 \quad \langle 0|\hat{a}^\dagger\hat{a}^\dagger|0\rangle = 0 \quad \hat{a}\,|0\rangle = 0$

$$\langle \hat{p}^2 \rangle = -\langle 0| \left[-\nu\mu\hat{a}\hat{a}^\dagger - \mu\nu\hat{a}\hat{a}^\dagger - 2\nu^2\hat{a}\hat{a}^\dagger - 1 \right] |0\rangle$$

$$\langle \hat{p}^2 \rangle = \nu\mu + \mu\nu + 2\nu^2 + 1$$

$\mu = \cosh r \quad \nu = \sinh r \quad \cosh^2 r - \sinh^2 r = 1$

$$\langle \hat{p}^2 \rangle = \nu\mu + \mu\nu - 2\nu^2 + \mu^2 - \nu^2$$

Squeezed vacuum state

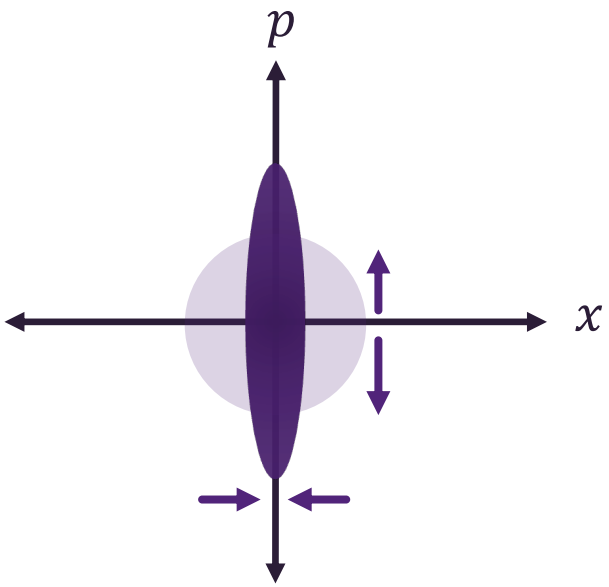
Variance

$$\langle \hat{p}^2 \rangle = (\mu + \nu)^2$$

$$\langle \hat{p}^2 \rangle = (\cosh r + \sinh r)^2$$

$$\langle \hat{p}^2 \rangle = e^{2r}$$

$$\mathrm{var}\,(p) = \langle \Delta \hat{p}^2 \rangle = \underbrace{\langle \hat{p}^2 \rangle}_{e^{2r}} - \underbrace{\langle \hat{p} \rangle^2}_{0} = e^{2r}$$



For the vacuum state: $\mathrm{var}\,(p) = 1$

After applying squeezing operator to vacuum: $\mathrm{var}\,(p) = e^{2r} > 1$

Squeezed vacuum state

Variance

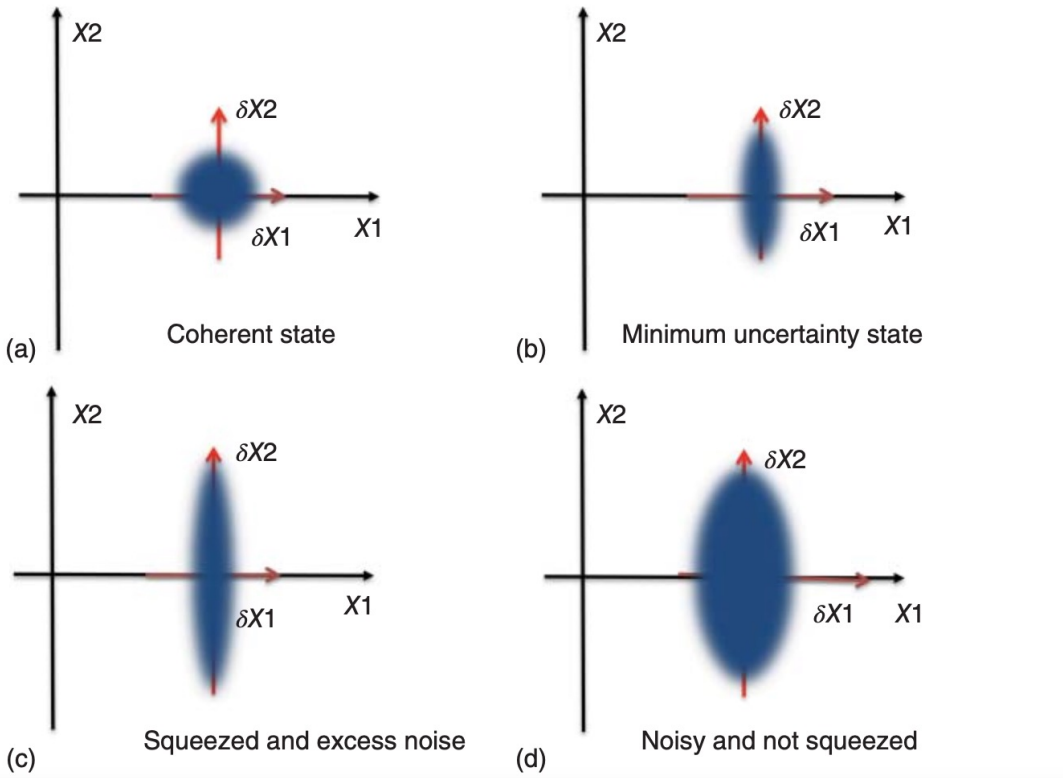
Uncertainty principle:

$$\langle \Delta \hat{x}^2 \rangle \langle \Delta \hat{p}^2 \rangle \geq 1$$

$$\mathrm{var}\,(x) = e^{-2r} \qquad \mathrm{var}\,(p) = e^{2r}$$

$$\langle \Delta \hat{x}^2 \rangle \langle \Delta \hat{p}^2 \rangle = e^{-2r} e^{2r} = 1$$

Squeezed states are minimum uncertainty states



p. 310 Bachor, Hans-A. / Ralph, Timothy C.
A guide to experiments in quantum optics (2019)

Generalised squeezed states

$$|\alpha, \zeta\rangle = \hat{D}(\alpha)\hat{S}(\zeta)\,|0\rangle$$

displacement operator

$$\hat{D}(\alpha) = e^{\alpha\hat{a}^\dagger - \alpha^*\hat{a}}$$

squeezing operator

$$S(\zeta) = \exp\left(\left(\zeta^*\hat{a}^2 - \zeta\hat{a}^{\dagger 2}\right)/2\right)$$

$$\zeta = re^{2i\phi}$$

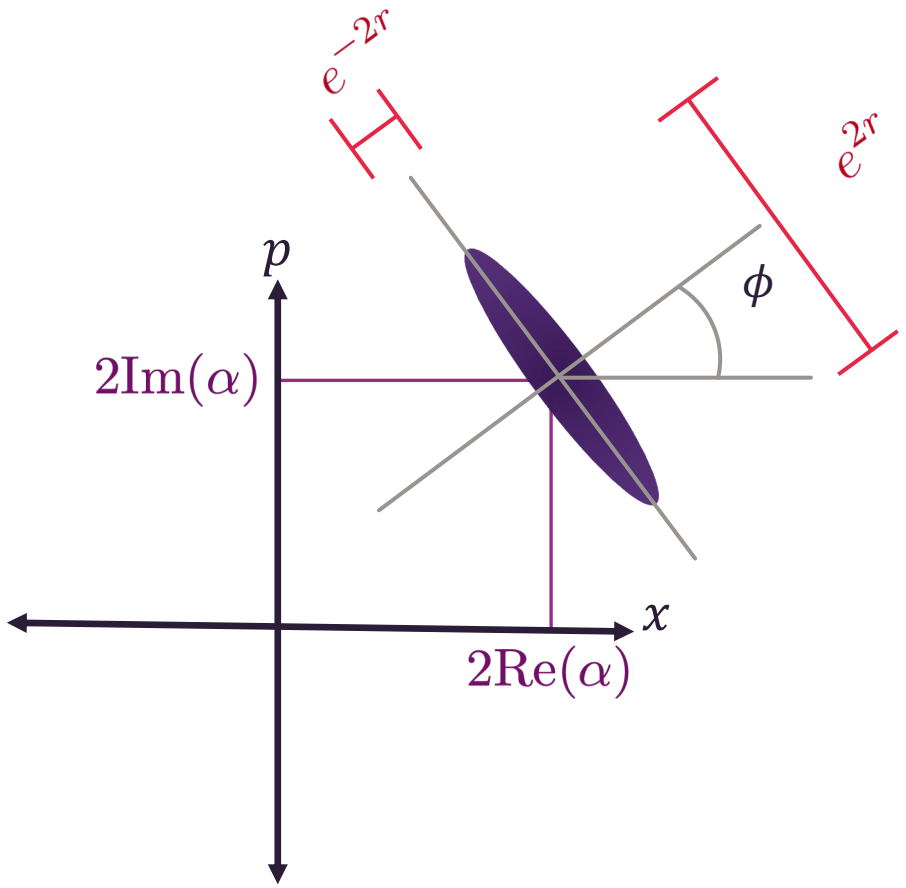
$$\langle\hat{x}\rangle = \alpha + \alpha^* = 2\mathrm{Re}(\alpha)$$

$$\langle\hat{p}\rangle = -i(\alpha^* - \alpha) = 2\mathrm{Im}(\alpha)$$

From the notes:

$$\langle\hat{x}^2\rangle = |\mu - \nu|^2 + (\alpha + \alpha^*)^2$$

$$\mu = \cosh r \qquad \nu = e^{2i\phi}\sinh r$$



Photon number distribution

In the photon number basis:

$$|0, r\rangle = S(r)\,|0\rangle$$
$$= \frac{1}{\sqrt{\cosh r}} \sum_{n=0}^{\infty} \frac{\sqrt{(2n)!}}{2^n n!} \tanh r^n\, |2n\rangle$$

squeezed vacuum only contains photon pairs

Probability of detecting a pair of photons

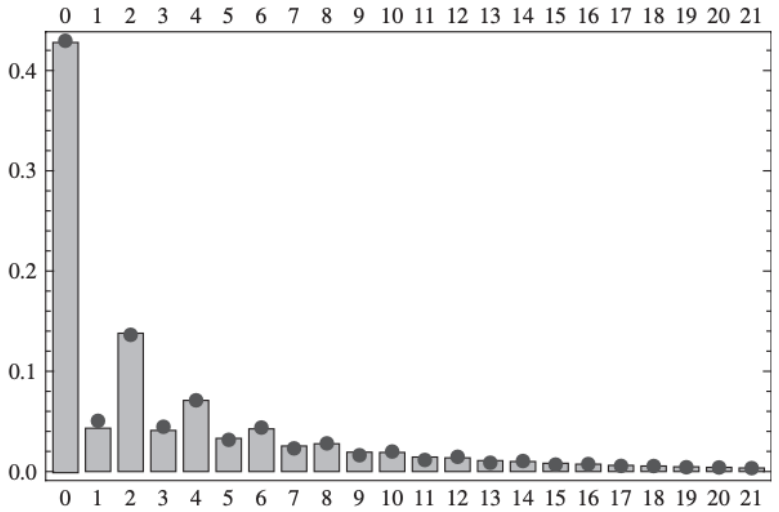
$$p_{2m} = \binom{2m}{m} \frac{1}{\cosh \zeta} \left(\frac{1}{2} \tanh \zeta\right)^{2m} \qquad (m = 0, 1, 2, \ldots)$$

Photons are created in pairs: each pump photon is converted into two signal photons of half the pump frequency

Due to detection inefficiencies sometimes only one photon of a pair is observed. This can be modelled with a beam splitter before the detector.

Odd photon numbers occur with non-zero probability

Photon number distribution of a squeezed vacuum.



p. 59 Leonhardt, U. Essential Quantum Optics From Quantum Measurements to Black Holes (2010)
(Data: Akira Furusawa and Hidehiro Yonezawa.)

How to generate squeezing

Squeezed light can be generated in nonlinear optical crystals.

→ Non linear interaction with the medium

$\chi^{(2)}$ non linearity
(LiNbO3 or lithium triborate (LBO))

Parametric amplification

interaction between three distinct light waves
(with different *frequencies*)

1) Pump
2) Signal

Pump photons are converted into signal and idler photons

Degenerate parametric amplification

$$H_{\chi^{(2)}} = \hbar\Omega a^\dagger a + 2\hbar\Omega c^\dagger c + \hbar\chi\left(c(a^\dagger)^2 + c^\dagger a^2\right)$$

Photon in the pump

Nonlinear interaction
one c-photon two a-photons

If the pump intensity is large: classical field

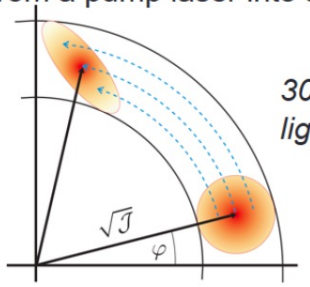
$$U(t) = \exp\left(-\frac{i}{\hbar}Ht\right) \propto S(\zeta)$$

vacuum input can generate a squeezed vacuum *PRL 100 (3), 033602 (2008)*

$\chi^{(3)}$ non linearity
(optical fiber)

four-wave-mixing (2 pumps now)
nonlinear optical *Kerr* effect

The Kerr effect transforms a coherent state from a pump laser into a squeezed state



30 years of squeezed light generation

Arxiv 1511.03250

In phase space, higher amplitude regions induce an increased phase shift

Significant pump powers required

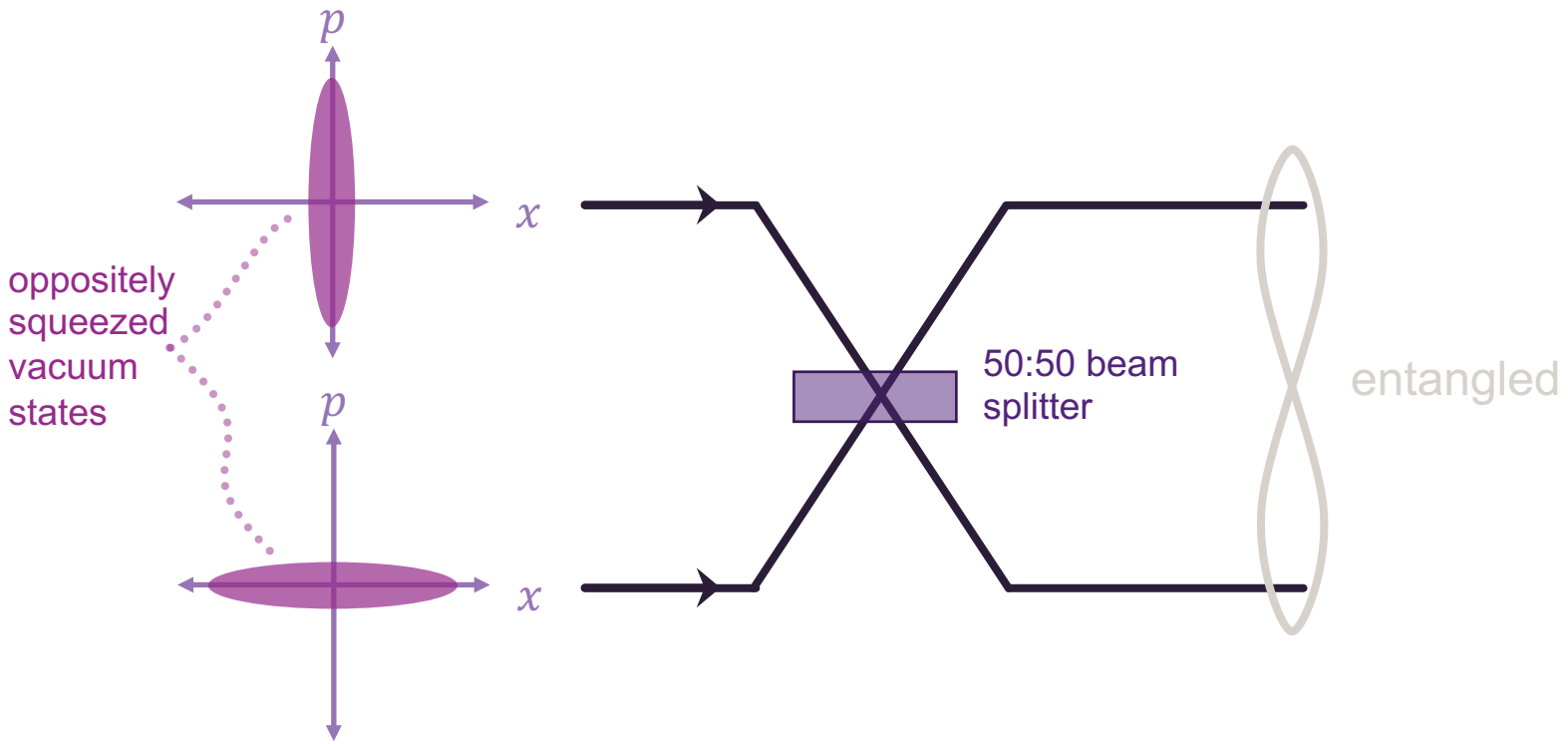
→ Phonon-photon coupling

Two-mode squeezed vacuum states

- If we combine two squeezed vacuum states, with opposite squeezing parameters, on a 50:50 beam splitter, the output will be the entangled two-mode squeezed vacuum state (TMSV)
- This state has quadrature correlations as well as photon number correlations

$$|TMSV\rangle = \sqrt{1-\lambda^2}\sum_{n}^{\infty}\lambda^n |n\rangle |n\rangle$$

$$\lambda = \tanh r$$



- This state is maximally entangled only in the limit as $\lambda \to \infty$ (but this limit is not physically achievable as it requires infinite energy)

Summary

What we talked about today…

The squeezing operator is defined as:

$$S(\zeta) = \exp\left(\left(\zeta^* \hat{a}^2 - \zeta \hat{a}^{\dagger 2}\right)/2\right)$$

Squeezed vacuum state:

$$|\zeta\rangle = \hat{S}(\zeta)\,|0\rangle$$

General squeezed state:

$$|\alpha, \zeta\rangle = \hat{D}(\alpha)\hat{S}(\zeta)\,|0\rangle$$

Squeezed states are minimum uncertainty states

$$\langle \Delta\hat{x}^2 \rangle \langle \Delta\hat{p}^2 \rangle = e^{-2r} e^{2r} = 1$$

